

Project 2: YSU Quantitative Finance

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- 1 Project 2b - Bayesian Update
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Bayesian Updates: Bernoulli Data

Model each binary outcome $X_i \in \{0, 1\}$ (e.g. is the daily return positive?) as $X_i \stackrel{iid}{\sim} \text{Bernoulli}(p)$.

Prior: $p \sim \text{Beta}(a, b)$, default $a = b = 1$ (Uniform)

Posterior (conjugacy), with $s = \sum x_i$ successes out of n :

$$p \mid \text{data} \sim \text{Beta}(a + s, b + n - s)$$

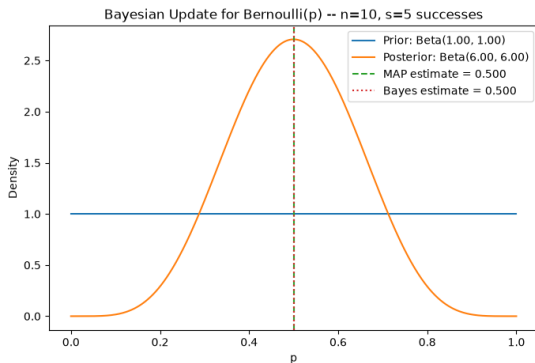
Estimates from posterior $\text{Beta}(a', b')$:

$$\hat{p}_{\text{MAP}} = \frac{a' - 1}{a' + b' - 2}, \quad \hat{p}_{\text{Bayes}} = \frac{a'}{a' + b'}$$

- **Why Beta?** Beta is the conjugate prior for Bernoulli because the posterior has the same Beta functional form as the prior – updating reduces to incrementing the shape parameters: $a' = a + s$, $b' = b + (n - s)$

Bernoulli Example: Toy Data

Prior Beta(1,1), data 1100101100 ($n = 10$, $s = 5$ successes)

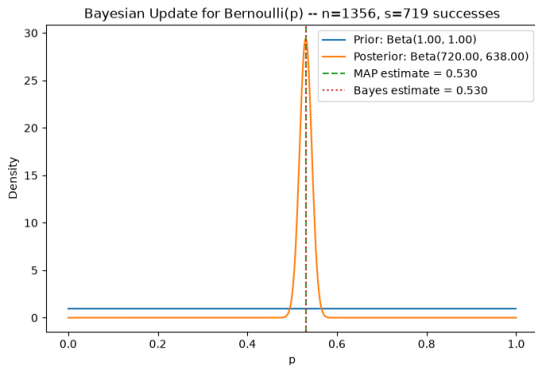


Posterior Beta(6, 6), $\hat{p}_{\text{MAP}} = \hat{p}_{\text{Bayes}} = 0.500$

- MAP and Bayes estimates coincide here because Beta(6, 6) is symmetric; for an asymmetric posterior they would generally differ

Bernoulli Example: Is AAPL's Daily Return Positive?

$X_t = 1$ if AAPL's daily log return is positive, 0 otherwise. Prior Beta(1, 1), $n = 1356$ days, $s = 719$ positive days.



Posterior Beta(720, 638), $\hat{p}_{\text{MAP}} = \hat{p}_{\text{Bayes}} \approx 0.530$

- AAPL closes up on $\approx 53\%$ of days – slightly more often than not
- 95% credible interval: [0.5036, 0.5567] – AAPL closes up slightly more than half the time, but with meaningful uncertainty remaining

Bayesian Updates: Normal Data

Model $X_i \stackrel{iid}{\sim} N(\mu, \sigma^2)$ with **known** σ (e.g. the mean daily return μ of a stock).

Prior: $\mu \sim N(\mu_0, \sigma_0^2)$, default $\mu_0 = 0$, $\sigma_0 = 1$

Posterior (conjugacy), with sample mean $\bar{x}$:

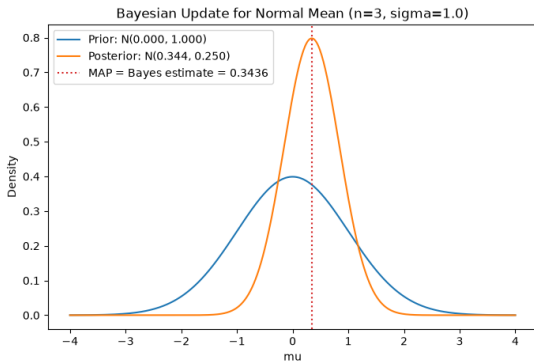
$$\mu \mid \text{data} \sim N(\mu_n, \sigma_n^2), \quad \frac{1}{\sigma_n^2} = \frac{1}{\sigma_0^2} + \frac{n}{\sigma^2}, \quad \mu_n = \sigma_n^2 \left(\frac{\mu_0}{\sigma_0^2} + \frac{n\bar{x}}{\sigma^2} \right)$$

For a Normal posterior, mode = mean:

$$\hat{\mu}_{\text{MAP}} = \hat{\mu}_{\text{Bayes}} = \mu_n$$

Normal Example: Toy Data

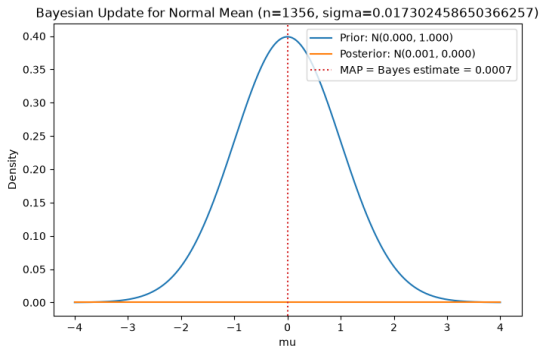
$\sigma = 1$, prior $N(0, 1)$, data 0.123, 1.3812, -0.13 ($n = 3$)



Posterior $N(0.344, 0.25)$, $\hat{\mu}_{\text{MAP}} = \hat{\mu}_{\text{Bayes}} = 0.344$

Normal Example: Mean of AAPL Daily Returns

σ fixed at the sample std of AAPL's log returns, prior $N(0, 1)$, $n = 1356$



Posterior mean $\hat{\mu}_{\text{Bayes}} \approx 0.0007$ – with $n = 1356$, the data completely dominates the (wide) prior

- 0.07% daily return compounds to approximately 19.28% annually (252 trading days) – statistically detectable but economically modest drift

Why Bayes? From Estimate to Decision

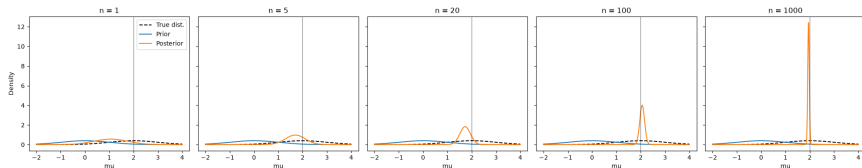
A frequentist gives a point estimate $\hat{p} = 0.530$ and a confidence interval. The Bayesian posterior gives the **full probability distribution** over p – so we can make direct probability statements:

- $P(p > 0.5 \mid \text{data}) = 0.987$ (98.7%)
- $P(p < 0.5 \mid \text{data}) = 0.013$ (1.3%)

We are 98.7% confident AAPL is an upward-drifting asset – a statement a frequentist confidence interval cannot make directly.

Supplementary: Does the Posterior Forget the Prior?

Known true $\mu = 2$, $\sigma = 1$; prior $N(0, 1)$ (deliberately off-center). Generate n points from the true distribution and compute the posterior.



- As n grows, the posterior narrows and shifts toward the true μ
- The prior's influence vanishes as $n \rightarrow \infty$
- **Bernstein–von Mises theorem:** as $n \rightarrow \infty$, the posterior converges to a Normal distribution centered at the true parameter, with variance equal to the inverse Fisher information – regardless of the prior
- **Implication:** with enough data, Bayesian and frequentist inference agree. The prior only matters when data is scarce

- **Selection:** stocks from one sector — **Technology** (5 names): AAPL, MSFT, NVDA, GOOGL, META
- **Market index:** ^GSPC (S&P 500)
- **Period:** 2021-01-01 to 2026-06-01, daily (≈ 1356 obs)
- **Variable:** daily log returns $r_t = \log(P_t/P_{t-1})$
- **Method:** OLS via `statsmodels`; inference re-run with heteroskedasticity / autocorrelation-robust (HC/HAC) standard errors

Strong Co-movement Across the Sector

Daily-return correlations among the five tech stocks and the index.



- All pairs are strongly positively correlated – the stocks move together
- This is the “can one stock explain another?” question: the answer is yes, but it reflects a **shared market factor**, not a direct link

Stage 1: Three Simple Linear Regressions

For each stock we fit three single-predictor models (shown here for AAPL):

(A) Market:
$$\text{AAPL}_t = \beta_0 + \beta_1 \text{S\&P 500}_t + \varepsilon_t$$

(B) Lagged market:
$$\text{AAPL}_t = \beta_0 + \beta_1 \text{S\&P 500}_{t-1} + \varepsilon_t$$

(C) Own lag (AR1):
$$\text{AAPL}_t = \beta_0 + \beta_1 \text{AAPL}_{t-1} + \varepsilon_t$$

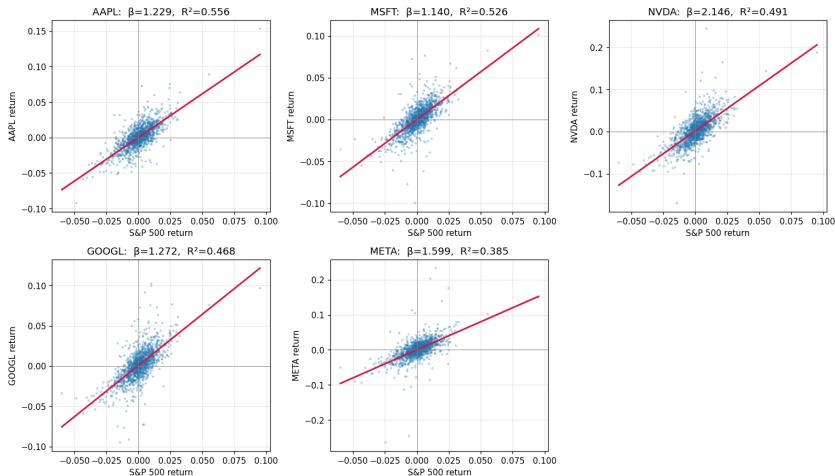
- (A) Can the market index explain individual stock returns?
- (B), (C) Can past returns predict future returns?

Model A: Market Model (contemporaneous)

Can market index returns explain individual stock returns?

$$\text{Stock}_t = \beta_0 + \beta_1 \cdot \text{S\&P 500}_t + \varepsilon_t$$

Model A — stock return vs contemporaneous market return



Model A: Results

	AAPL	MSFT	NVDA	GOOGL	META
β (market)	1.229	1.140	2.146	1.272	1.599
R^2	0.556	0.526	0.491	0.468	0.385

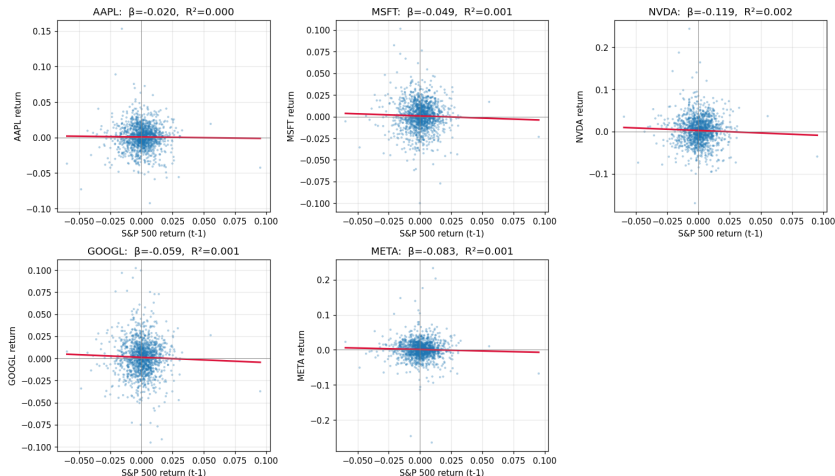
- All five betas are large and **highly significant** ($p \approx 0$); β is the stock's **market beta** (systematic-risk loading)
- Every $\beta > 1$ (NVDA highest at 2.15, MSFT lowest at 1.14) – these tech names amplify market moves
- $R^2 \approx 0.39$ –0.56: the market explains a large share of each stock's daily variance; the rest ($1 - R^2$) is idiosyncratic
- Intercept α (Jensen's alpha) is **not** significant – no abnormal return once market risk is accounted for
- Residuals are fat-tailed (heavy tails in the Q–Q plot) – typical of daily returns

Model B: Lagged Market $\rightarrow$ Stock Return

Can yesterday's market move predict today's return?

$$\text{Stock}_t = \beta_0 + \beta_1 \cdot \text{S\&P 500}_{t-1} + \varepsilon_t$$

Model B — stock return vs LAGGED market return

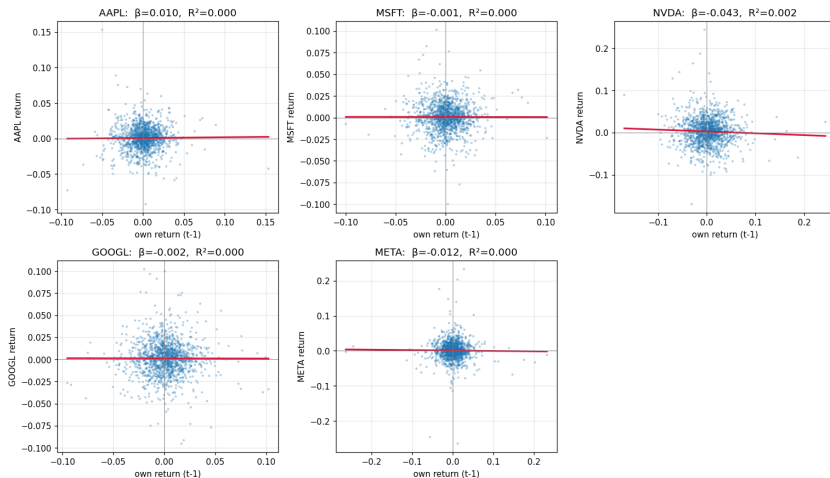


Model C: Own Lagged Return (AR1)

Does a stock's own past return predict its next return?

$$\text{Stock}_t = \beta_0 + \beta_1 \cdot \text{Stock}_{t-1} + \varepsilon_t$$

Model C — stock return vs its own lagged return (AR(1))



Stage 1: Summary

Model	Predictor	R^2	Significant?
A	S&P 500 _t	0.556	Yes
B	S&P 500 _{t-1}	≈ 0.000	No
C	AAPL _{t-1}	≈ 0.000	No

(R^2 shown for AAPL; full sector range in Model A is 0.39–0.56.)

- Same-day market index return strongly explains each stock's return ($R^2 \approx 0.39$ –0.56 across the sector)
- Lagged (past) returns – index or own – carry **no** predictive information: consistent with weak-form market efficiency
- Cross-stock co-movement (e.g. GOOGL vs AAPL) is really shared exposure to this same market factor

Stage 2: Multiple Linear Regression

Predict a target stock's return from **several predictors jointly**:

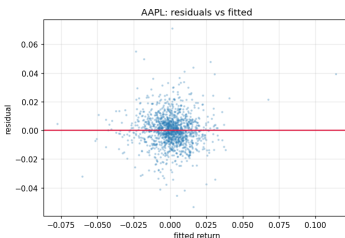
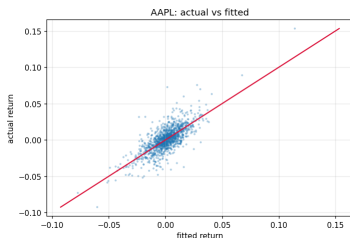
$$\begin{aligned} \text{AAPL}_t = & \beta_0 + \beta_1 \text{S\&P } 500_t + \beta_2 \text{S\&P } 500_{t-1} + \beta_3 \text{AAPL}_{t-1} + \beta_4 \text{AAPL}_{t-2} \\ & + \beta_5 \text{MAGap}_{t-1} + \beta_6 \text{Vol}_{t-1} + \beta_7 \text{RelVol}_{t-1} + \varepsilon_t \end{aligned}$$

- MAGap: price vs its 20-day moving average; Vol: 20-day realised volatility; RelVol: volume vs its 20-day average
- All predictors except the same-day market are **strictly lagged** – no look-ahead, so a model without $\text{S\&P } 500_t$ is genuinely usable to forecast

Stage 2: Results (AAPL full model)

Obtained model (only the significant term shown explicitly):

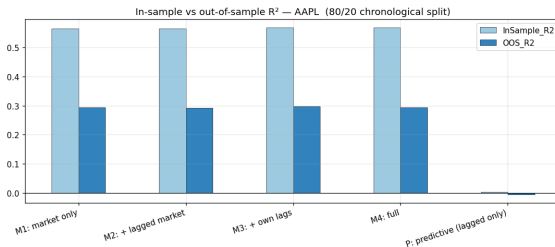
$$\widehat{\text{AAPL}}_t \approx \hat{\beta}_0 + 1.23 \text{ S\&P } 500_t + (6 \text{ small, insignificant terms})$$



- Only the **same-day market** term is significant: $\hat{\beta}_1 \approx 1.23$, $p < 0.001$ (matches the Stage-1 beta); the six lagged/technical terms are individually insignificant ($p > 0.05$)
- $R^2 \approx 0.57$, adjusted $R^2 \approx 0.56$ – barely above the market-only model; lagged & technical predictors add almost nothing
- All VIFs < 5 – no multicollinearity; the predictors simply carry no signal

Stage 3: Model Selection (in- vs out-of-sample)

Candidate models fit on the **same sample**; out-of-sample uses a chronological 80/20 split, benchmarked against the historical-mean forecast.

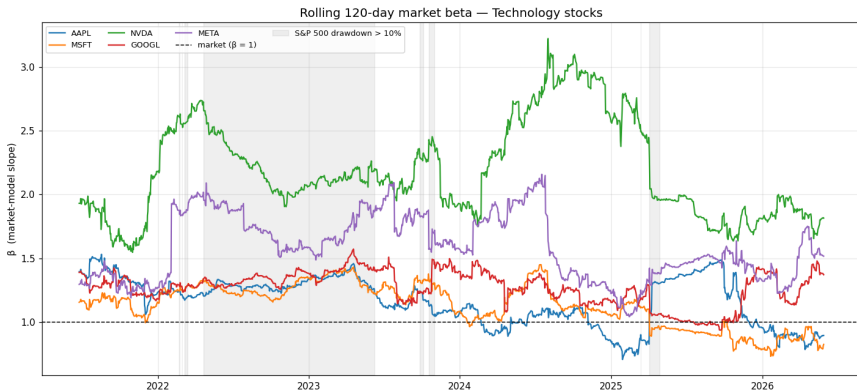


Model	adj. R^2	R^2_{Oos}
Market-only (M1)	≈ 0.56	≈ 0.29
Full, 7 predictors (M4)	≈ 0.56	≈ 0.29
Predictive, lagged-only (P)	≈ 0.00	≈ 0.00

- AIC/BIC and adjusted R^2 prefer the parsimonious **market-only** model
- The honest lagged-only model has $R^2_{\text{Oos}} \approx 0$ (slightly negative): **past**

Extension: Beta Is Not Constant

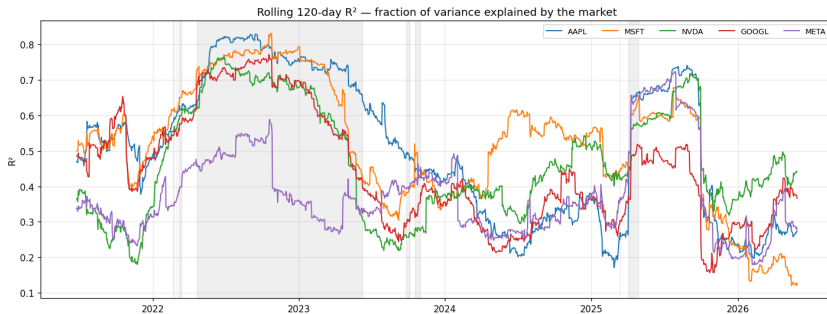
Re-estimate the market model on a rolling 120-day window: $\hat{\beta}_t$ over time.



- The single Stage-1 beta is a 5-year *average* – the true loading drifts a lot
- Betas climb inside market drawdowns (shaded): in selloffs correlations spike and **everything moves together** – risk is state-dependent

Extension: Systematic Share Over Time

Rolling R^2 of the market model – the fraction of variance the market explains.



- R^2 rises in stressed periods – the market dominates exactly when diversification is least helpful

Linear Regression: Summary

- **Market explains, contemporaneously:** same-day index return is the dominant driver ($R^2 \approx 0.39\text{--}0.56$); its slope is the stock's beta
- **Past returns do not predict:** lagged index, own lags, moving-average, volatility and volume are jointly insignificant and fail out-of-sample
- **Volume:** does not explain the *signed* daily return
- **Best model** = the market model; **best genuine predictor** = the mean
- **Beta is dynamic:** it rises in drawdowns – a static regression hides the risk exactly when it matters most

Defining Market Regimes

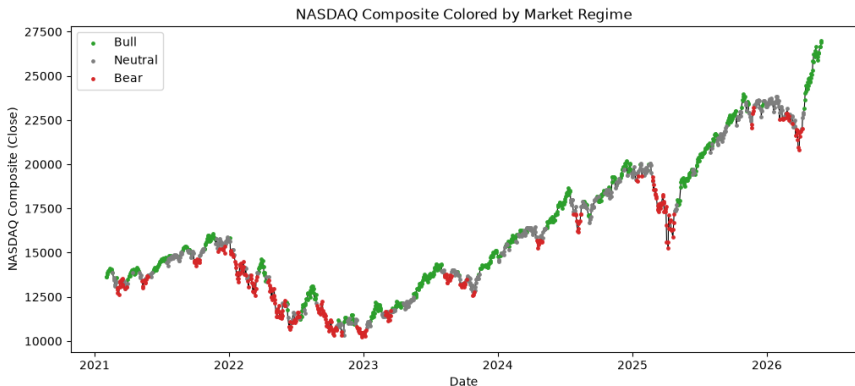
Classify each day's NASDAQ Composite ($\hat{IXIC}$) into **Bull**, **Neutral**, or **Bear** using the 20-day rolling cumulative log return

$$R_t = \sum_{i=0}^{19} r_{t-i}, \quad \theta = 0.5 \cdot \text{std}(R_t)$$

$$\text{Regime}_t = \begin{cases} \text{Bull} & R_t > \theta \\ \text{Bear} & R_t < -\theta \\ \text{Neutral} & \text{otherwise} \end{cases}$$

Regime	Bull	Neutral	Bear
Days (%)	554 (41.4%)	465 (34.8%)	318 (23.8%)

NASDAQ Composite Colored by Regime



Transition Probability Matrix

$$P_{ij} = P(\text{Regime}_{t+1} = j \mid \text{Regime}_t = i)$$

$$P = \begin{pmatrix} 0.879 & 0.121 & 0.000 \\ 0.144 & 0.751 & 0.105 \\ 0.000 & 0.154 & 0.846 \end{pmatrix} \quad (\text{rows/cols: Bull, Neutral, Bear})$$

- All three regimes are highly **persistent** (diagonal ≈ 0.75 – 0.88)
- $P(\text{Bull} \rightarrow \text{Bear}) = P(\text{Bear} \rightarrow \text{Bull}) = 0$ – the market always passes through Neutral

Stationary Distribution

$$\pi P = \pi, \sum_i \pi_i = 1$$

Regime	Stationary π	Observed frequency
Bull	0.414	0.414
Neutral	0.348	0.348
Bear	0.238	0.238

- Stationary distribution matches observed frequencies almost exactly – the chain is essentially already at its long-run distribution

Predicting the Next Regime

Given the current regime i , the most probable next regime is $\arg \max_j P_{ij}$.

- Last observed regime: **Bull**
- $P(\text{Bull} \rightarrow \text{Bull}) = 0.879$, $P(\text{Bull} \rightarrow \text{Neutral}) = 0.121$,
 $P(\text{Bull} \rightarrow \text{Bear}) = 0.000$
- Predicted next regime: **Bull**

Simulating the Markov Chain

Simulate a regime sequence of the same length from P and compare to the observed data.

Regime	Frequency	
	Observed	Simulated
Bull	0.414	0.476
Neutral	0.348	0.301
Bear	0.238	0.223

Regime	Avg. run length (days)	
	Observed	Simulated
Bull	8.1	8.8
Neutral	4.0	3.3
Bear	6.5	6.0

- A first-order Markov chain reproduces both the regime frequencies and the average regime durations reasonably well

- Market regimes (Bull/Neutral/Bear) are **highly persistent**, and the chain is essentially at its stationary distribution
- The model predicts the current Bull regime will most likely continue
- However, persistence is largely **mechanical**: R_t is a 20-day rolling sum, so R_t and R_{t-1} share 19 of 20 terms and are highly correlated by construction
- Consistent with Stage 1: *daily returns* themselves remain essentially unpredictable (weak-form efficiency); only the smoothed regime label is persistent

Questions?

Thank you!